

Forest Queries II

Time limit: 1.00 s

Memory limit: 512 MB

You are given an $n \times n$ grid representing the map of a forest. Each square is either empty or has a tree. Your task is to process q queries of the following types:

1. Change the state (empty/tree) of a square.
2. How many trees are inside a rectangle in the forest?

Input

The first input line has two integers n and q : the size of the forest and the number of queries. Then, there are n lines describing the forest. Each line has n characters: `.` is an empty square and `*` is a tree. Finally, there are q lines describing the queries. The format of each line is either `"1 y x"` or `"2 y1 x1 y2 x2"`.

Output

Print the answer to each query of the second type.

Constraints

- $1 \leq n \leq 1000$
- $1 \leq q \leq 2 \cdot 10^5$
- $1 \leq y, x \leq n$
- $1 \leq y_1 \leq y_2 \leq n$
- $1 \leq x_1 \leq x_2 \leq n$

Example

Input	Output
4 3 .*.. *.. **.. **** 2 2 2 3 4 1 3 3 2 2 2 3 4	3 4